

12 Real-World ML Project Themes

Each theme = 1 Problem Statement + 4 ML Models to Compare

THEMES



12 Domains

PER THEME



4 ML Models

TOTAL



48 Models

OUTPUT



Comparison Report

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ALL 12 THEMES



Pick Your Battle

THEME 1

 **Healthcare**

Diabetes Prediction

THEME 2

 **Real Estate**

House Price Prediction

THEME 3

 **Banking**

Loan Default Prediction

THEME 4

 **E-Commerce**

Customer Churn

THEME 5

 **HR Analytics**

Employee Attrition

THEME 6

 **Insurance**

Premium Prediction

THEME 7

 **Education**

Student Performance

THEME 8

 **Agriculture**

Crop Yield Prediction

THEME 9

 **Automotive**

Used Car Valuation

THEME 10

 **Telecom**

Network Churn

THEME 11

 **Food & Restaurant**

Rating Prediction

THEME 12

 **Sports Analytics**

IPL Player Score

Every project requires: Linear / Logistic Reg Polynomial Reg Decision Tree Random Forest → Compare all 4 and declare a winner!



Healthcare — Diabetes Prediction

 PROBLEM STATEMENT

Build an ML system that predicts whether a patient is **diabetic or not** based on medical diagnostic measurements (glucose level, blood pressure, BMI, age, insulin, etc.). Train 4 models, compare accuracy, precision & recall, and recommend the best model for hospital deployment.

 DATASET

Pima Indians Diabetes Dataset — 768 patients, 8 features, binary target (0/1). Available on Kaggle.

 Target Variable

Outcome → **0** = Not Diabetic, **1** = Diabetic (Classification)

 4 Models to Build & Compare

#	Model	Why This Model?
M1	Logistic Regression	Baseline classifier, interpretable
M2	Polynomial Logistic (deg=2)	Capture non-linear health patterns
M3	Decision Tree Classifier	Easy to explain to doctors
M4	Random Forest Classifier	Best accuracy, handles noise

 Deliverables

- EDA with correlation heatmap
- Feature Importance bar chart
- Accuracy, Precision, Recall, F1 comparison table
- Confusion Matrix for best model
- Final recommendation with reasoning



Real Estate — House Price Prediction

 **PROBLEM STATEMENT**
Build an ML system that predicts the **selling price of houses** in Bengaluru/Boston based on features like area, bedrooms, location, age, and amenities. Compare 4 regression models using MAE, RMSE, and R^2 to find the best price predictor.

 **DATASET**
Bengaluru House Price / Boston Housing — 13,000+ rows, 9 features. Available on Kaggle.

 **Target Variable**
price → Continuous value in ₹ Lakhs (Regression)

 4 Models to Build & Compare		
#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline, check linear fit
M2	Polynomial Regression (deg=2)	Capture curved pricing patterns
M3	Decision Tree Regressor	Non-linear splits, interpretable
M4	Random Forest Regressor	Best accuracy, handles outliers

-  **Deliverables**
- Location-wise price distribution plots
 - Scatter plot: Predicted vs Actual prices
 - MAE, RMSE, R^2 comparison table
 - Feature importance (which factor drives price most?)
 - Final model recommendation



Banking — Loan Default Prediction

PROBLEM STATEMENT

Build an ML system for a bank to predict whether a loan applicant will **default (not repay)** based on income, credit history, employment, loan amount, and property area. Compare 4 classifiers to find the most reliable model for risk assessment.

DATASET

Loan Prediction Dataset — 614 rows, 12 features, binary target.
Kaggle / Analytics Vidhya.

Target Variable

Loan_Status → **Y** = Approved, **N** = Rejected (Classification)

4 Models to Build & Compare

#	Model	Why This Model?
M1	Logistic Regression	Industry standard for credit scoring
M2	Polynomial Logistic (deg=2)	Non-linear income-risk relationship
M3	Decision Tree Classifier	Explainable to regulators
M4	Random Forest Classifier	Handles missing data, best accuracy

Deliverables

- Income vs Loan Amount scatter analysis
- Credit History impact analysis
- ROC-AUC curve comparison
- Confusion Matrix (False Negatives critical!)
- Business recommendation report



E-Commerce — Customer Churn Prediction

PROBLEM STATEMENT

An e-commerce company is losing customers. Build an ML system to predict which customers will **stop buying (churn)** based on order frequency, complaints, cashback, tenure, and preferred payment mode. Identify the top 3 features driving churn.

DATASET

E-Commerce Customer Churn — 5,630 rows, 20 features, binary target. Kaggle.



Target Variable

Churn → **0** = Stayed, **1** = Left (Classification)



4 Models to Build & Compare

#	Model	Why This Model?
M1	Logistic Regression	Quick baseline, odds ratios
M2	Polynomial Logistic (deg=2)	Capture tenure-churn curves
M3	Decision Tree Classifier	Visualize churn decision path
M4	Random Forest Classifier	Best for high-dimensional data



Deliverables

- Churn rate by payment mode / category
- Feature Importance: Top 3 churn drivers
- Precision-Recall tradeoff analysis
- Cost-benefit analysis (cost of losing a customer)
- Retention strategy recommendations



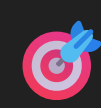
HR Analytics — Employee Attrition Prediction

 **PROBLEM STATEMENT**

A company's HR team wants to predict which employees are likely to **resign** based on satisfaction level, salary, overtime, promotions, and distance from home. Build 4 models and identify what makes employees leave.

 **DATASET**

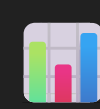
IBM HR Analytics Attrition — 1,470 rows, 35 features, binary target. Kaggle.

 **Target Variable**

Attrition → **Yes** = Left, **No** = Stayed (Classification)

 **4 Models to Build & Compare**

#	Model	Why This Model?
M1	Logistic Regression	Identify risk factors (odds ratios)
M2	Polynomial Logistic (deg=2)	Non-linear satisfaction curves
M3	Decision Tree Classifier	Visual rule: "If overtime=Yes AND..."
M4	Random Forest Classifier	Handle 35 features, best accuracy

 **Deliverables**

- Attrition rate by department & role
- Satisfaction vs Attrition correlation
- Decision Tree visualization (export as image)
- Top 5 attrition drivers (Feature Importance)
- HR policy recommendations

Insurance — Premium Cost Prediction

 **PROBLEM STATEMENT**
An insurance company needs to predict the **annual medical insurance premium** a customer should be charged based on age, BMI, smoking status, number of children, and region. Build 4 regression models and recommend the most accurate pricing engine.

 **DATASET**
Medical Cost Personal Dataset — 1,338 rows, 6 features, continuous target. Kaggle.

 **Target Variable**
charges → Continuous value in \$ (Regression)

 4 Models to Build & Compare		
#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline pricing model
M2	Polynomial Regression (deg=2)	Age-cost curve is non-linear
M3	Decision Tree Regressor	Smoker vs non-smoker splits
M4	Random Forest Regressor	Best accuracy, handles interactions

-  **Deliverables**
- Smoker vs Non-smoker cost distribution
 - Age vs Charges scatter (show curve)
 - MAE, RMSE, R² comparison table
 - Predicted vs Actual scatter plot
 - Pricing model recommendation



Education — Student Performance Prediction

 **PROBLEM STATEMENT**

A school wants to predict students' **final exam scores** based on study hours, attendance, parent education, internet access, and past failures. Build 4 regression models to identify at-risk students early for intervention.

 **DATASET**

Student Performance Dataset (UCI) — 649 rows, 30 features, continuous target. Kaggle / UCI.

 **Target Variable**

G3 → Final grade (0–20 scale) (Regression)

 **4 Models to Build & Compare**

#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline score predictor
M2	Polynomial Regression (deg=2)	Diminishing returns of study hours
M3	Decision Tree Regressor	"If failures > 2 AND study < 3h..."
M4	Random Forest Regressor	Handle 30 mixed features

-  **Deliverables**
- Study hours vs Final grade scatter
 - Attendance impact analysis
 - Parent education correlation
 - At-risk student identification system
 - Intervention recommendations for school



Agriculture — Crop Yield Prediction

 PROBLEM STATEMENT

Build an ML system for Indian farmers to predict **crop yield (tons/hectare)** based on rainfall, temperature, soil type, fertilizer usage, and pesticide usage. Help farmers make data-driven planting decisions.

 DATASET

Crop Yield Prediction Dataset — 28,000+ rows, 7 features, continuous target. Kaggle.

 Target Variable

yield → Tons per hectare (Regression)

 4 Models to Build & Compare

#	Model	Why This Model?
M1	Multiple Linear Regression	Simple baseline, quick estimate
M2	Polynomial Regression (deg=2)	Rainfall-yield curve (too much rain = bad)
M3	Decision Tree Regressor	Region-specific rules
M4	Random Forest Regressor	Best for environmental data

 Deliverables

- Rainfall vs Yield curve (show polynomial fit)
- Crop-wise yield comparison
- Fertilizer impact analysis
- State/region performance heatmap
- Farmer advisory dashboard concept

Automotive — Used Car Price Prediction

PROBLEM STATEMENT

Build a **used car valuation engine** (like Cars24/OLX) that predicts resale price based on brand, year, km driven, fuel type, seller type, and transmission. Help sellers price their cars competitively.

DATASET

Vehicle Dataset (CarDekho) — 8,100+ rows, 13 features, continuous target. Kaggle.

Target Variable

selling_price → Price in ₹ Lakhs (Regression)

4 Models to Build & Compare

#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline depreciation model
M2	Polynomial Regression (deg=2)	Depreciation curve (steep then flat)
M3	Decision Tree Regressor	Brand-specific pricing rules
M4	Random Forest Regressor	Handle brand + age + km interactions

Deliverables

- Year vs Price depreciation curve
- Brand-wise average resale value
- Fuel type impact on pricing
- KM driven vs Price scatter with trend line
- Price calculator demo



Telecom — Customer Churn Prediction

 PROBLEM STATEMENT

A telecom company (like Jio/Airtel) wants to predict which customers will **switch to a competitor** based on contract type, monthly charges, tenure, tech support calls, and internet service type. Reduce churn by targeting at-risk users.

 DATASET

Telco Customer Churn — 7,043 rows, 21 features, binary target. Kaggle (IBM).

 Target Variable

Churn → **Yes** = Left, **No** = Stayed (Classification)

 4 Models to Build & Compare

#	Model	Why This Model?
M1	Logistic Regression	Baseline, identify risk factors
M2	Polynomial Logistic (deg=2)	Tenure-churn non-linear pattern
M3	Decision Tree Classifier	Rule-based churn detection
M4	Random Forest Classifier	Handle 21 mixed-type features

 Deliverables

- Churn rate by contract type (monthly vs yearly)
- Monthly charges distribution (churned vs stayed)
- Tenure vs Churn probability curve
- Top 5 churn predictors
- Retention offer strategy



Food & Restaurant — Rating Prediction

PROBLEM STATEMENT

Build an ML system to predict a restaurant's **Zomato/Swiggy aggregate rating** based on location, cuisine type, cost for two, online ordering availability, table booking, and number of votes. Help new restaurants estimate their expected rating.

DATASET

Zomato Bangalore Restaurants — 51,000+ rows, 17 features, continuous target. Kaggle.

Target Variable

rate → Rating on 5-point scale (Regression)

4 Models to Build & Compare

#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline, check linear factors
M2	Polynomial Regression (deg=2)	Cost-rating curve (diminishing returns)
M3	Decision Tree Regressor	Location + cuisine interaction rules
M4	Random Forest Regressor	Handle 17 features, text-heavy data

Deliverables

- Location-wise average rating heatmap
- Cost for two vs Rating scatter
- Online ordering impact analysis
- Top cuisines by rating
- New restaurant rating estimator



Sports Analytics — IPL Player Score Prediction

PROBLEM STATEMENT

Build an ML system to predict an IPL batsman's **runs scored in a match** based on batting average, strike rate, venue, opposition bowling strength, toss result, and batting position. Help fantasy cricket players make data-driven picks!

DATASET

IPL Complete Dataset (2008–2024) — 200,000+ deliveries, derived player stats. Kaggle.



Target Variable

runs_scored → Total runs by batsman in a match (Regression)



4 Models to Build & Compare

#	Model	Why This Model?
M1	Multiple Linear Regression	Baseline performance predictor
M2	Polynomial Regression (deg=2)	Form curves (peak → decline)
M3	Decision Tree Regressor	Venue + position specific rules
M4	Random Forest Regressor	Best for multi-factor sports data



Deliverables

- Player performance by venue analysis
- Strike rate vs Runs correlation
- Batting position impact on scoring
- Top predictive features for run scoring
- Fantasy XI recommendation engine



What to Submit

Project Structure

```
your_project/
├── data/
│   └── dataset.csv
├── notebooks/
│   └── analysis.ipynb
├── models/
│   ├── model_1_logistic.pkl
│   ├── model_2_polynomial.pkl
│   ├── model_3_decision_tree.pkl
│   └── model_4_random_forest.pkl
├── results/
│   ├── comparison_table.png
│   └── feature_importance.png
├── README.md
└── presentation.pdf
```

✓ Mandatory Checklist

- ☐ EDA with at least 5 visualizations
- ☐ Data cleaning (handle nulls, outliers)
- ☐ Feature engineering (at least 1 new feature)
- ☐ Train-Test Split (80:20)
- ☐ All 4 models trained & evaluated
- ☐ Comparison table (Accuracy/ R^2 , MAE/RMSE)
- ☐ Feature Importance chart
- ☐ Best model justification (2-3 sentences)
- ☐ GitHub repository link
- ☐ 5-minute presentation

★ Bonus Points

- Hyperparameter tuning (GridSearchCV)
- Cross-validation (5-fold)
- Streamlit/Gradio web app demo

LET'S BUILD!

12 Themes. 48 Models. 1 Portfolio That Gets You Hired.

"Don't just learn ML — **prove it** with a project that solves a real problem."



Healthcare



Real Estate



Banking



E-Commerce



HR Analytics



Insurance



Education



Agriculture



Automotive



Telecom



Food &
Restaurant



Sports
Analytics

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